

```

# =====
# 17. Noise Removal Comparison
# Google Colab Implementation
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# -----
# STEP 1: Upload Image
# -----
print("Upload an image (jpg/png)")
uploaded = files.upload()

# Get uploaded file name
img_path = next(iter(uploaded))

# -----
# STEP 2: Read Image
# -----
img = cv2.imread(img_path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# STEP 3: Add Gaussian Noise
# -----
noise = np.random.normal(0, 25, img.shape).astype(np.uint8)
noisy_img = cv2.add(img, noise)

# -----
# STEP 4: Apply Noise Removal Filters
# -----

# 1. Gaussian Filter
gaussian = cv2.GaussianBlur(noisy_img, (5, 5), 0)

# 2. Median Filter
median = cv2.medianBlur(noisy_img, 5)

```

```
# 3. Bilateral Filter
bilateral = cv2.bilateralFilter(noisy_img, 9, 75, 75)

# -----
# STEP 5: Display Results
# -----
images = [img, noisy_img, gaussian, median, bilateral]
titles = [
    'Original Image',
    'Noisy Image',
    'Gaussian Filter',
    'Median Filter',
    'Bilateral Filter'
]

plt.figure(figsize=(18, 8))

for i in range(len(images)):
    plt.subplot(1, 5, i + 1)
    plt.imshow(images[i])
    plt.title(titles[i], fontsize=11)
    plt.axis('off')

plt.suptitle('Noise Removal Comparison', fontsize=18, fontweight='bold')
plt.tight_layout()
plt.show()
```

Upload an image (jpg/png)

img6.png

img6.png(image/png) - 3045754 bytes, last modified: 10/08/2026 - 100% done

Saving img6.png to img6.png

Noise Removal Comparison

Original Image



Noisy Image



Gaussian Filter



Median Filter



Bilateral Filter

